

L-Tyrosine Catabolism to Fumarate and Acetoacetate: A Mechanistic Review

A commissioned review-style synthesis of the five-reaction module that funnels L-tyrosine into central carbon metabolism.

1. Executive Summary

L-Tyrosine catabolism to fumarate and acetoacetate is a compact, linear, five-enzyme metabolic module whose function is to dismantle the aromatic ring of an amino acid and deliver its carbon skeleton into central metabolism as one glucogenic product (fumarate) and one ketogenic product (acetoacetate). The route proceeds in an obligatory chemical order: (1) transamination of L-tyrosine to 4-hydroxyphenylpyruvate by tyrosine aminotransferase (TAT) or the bacterial aromatic aminotransferase TyrB; (2) oxidative decarboxylation and ring hydroxylation to homogentisate by 4-hydroxyphenylpyruvate dioxygenase (HPD/HPPD); (3) oxidative aromatic ring cleavage to maleylacetoacetate by homogentisate 1,2-dioxygenase (HGD); (4) *cis*→*trans* isomerization to fumarylacetoacetate by the glutathione-dependent maleylacetoacetate isomerase (GSTZ1/MAAI); and (5) terminal hydrolysis to fumarate + acetoacetate by fumarylacetoacetate hydrolase (FAH). The pathway was reconstituted in its entirety in *E. coli* from cloned plant enzymes, confirming that these activities alone constitute a complete functional route from homogentisate to the two central-metabolism products (Dixon & Edwards, [PMID: 22980205](#); Han et al., [PMID: 23743712](#)).

A central and perhaps under-appreciated point emerging from this synthesis is that the module is **not** the product of a single enzyme family. It is a **mosaic of at least four structurally unrelated enzyme folds** — a PLP-dependent aminotransferase, two mechanistically distinct non-heme Fe(II) dioxygenases, a glutathione-transferase-zeta isomerase, and a member of the ancient, largely bacterial FAH hydrolase superfamily — recruited and stitched together to accomplish one chemical goal. Much of the deep enzymology (the FAH fold, the homogentisate convergence point) is bacterially rooted, while lineage-specific elaborations (glucocorticoid-regulated TAT in mammalian liver, homogentisate diversion to tocopherol/plastoquinone in plants) represent later specializations.

The boundaries of the system are drawn with unusual clarity by human inborn errors of metabolism, each of which maps to loss of a single step: **HPD deficiency → tyrosinemia type III / hawkinsinuria**; **HGD deficiency → alkaptonuria**; **FAH deficiency → hereditary tyrosinemia type 1 (HT1)**. The clinical severity gradient — mild at the top, lethal at the bottom — is explained mechanistically: the further downstream the block, the more chemically reactive the accumulating intermediate. Fumarylacetoacetate and its derivative succinylacetone are potent, non-competitive inhibitors of δ -aminolevulinic acid dehydratase (ALAD/porphobilinogen synthase), producing the porphyria-like neurovisceral crises of HT1. This mechanistic insight is also the therapeutic rationale for nitisinone (NTBC), which blocks the *upstream* enzyme HPD to prevent formation of the toxic downstream species.

2. Definition and Biological Boundaries

2.1 What is included

The system, as scoped here, is the **lower tyrosine catabolic trunk**: the ordered set of five reactions converting L-tyrosine → 4-hydroxyphenylpyruvate → homogentisate → maleylacetoacetate → fumarylacetoacetate → (fumarate + acetoacetate). This is a self-contained "funnel": once carbon enters as homogentisate, it is committed to fumarate + acetoacetate. Reconstitution of the three lower enzymes (HGO, MAAI, FAH) from *Arabidopsis* in a heterologous host produced a "fully functional catabolic pathway when combined in vitro" (Dixon & Edwards, [PMID: 22980205](#)), establishing that no additional accessory factors are strictly required for turnover.

2.2 Neighboring processes that should be treated separately

- **Upstream phenylalanine input.** Phenylalanine hydroxylase (PAH) converts phenylalanine to tyrosine and thereby *feeds* the module, but it is not part of it. In hibernating bats, PAH is co-upregulated with HGD and FAH as a coordinated catabolic response ([PMID: 23620802](#)), but PAH belongs to the aromatic-amino-acid-hydroxylase family and is upstream of the transamination entry step.
- **Alternative aromatic entry routes into the homogentisate node.** In bacteria such as *Pseudomonas putida* CSV86, the homogentisate pathway is a **convergence point** that also receives carbon from phenylacetate, *p*-hydroxyphenylacetate, and phenylpropanoids

([PMID: 24475028](#)). These peripheral entry routes share the lower trunk but are outside the tyrosine-specific module.

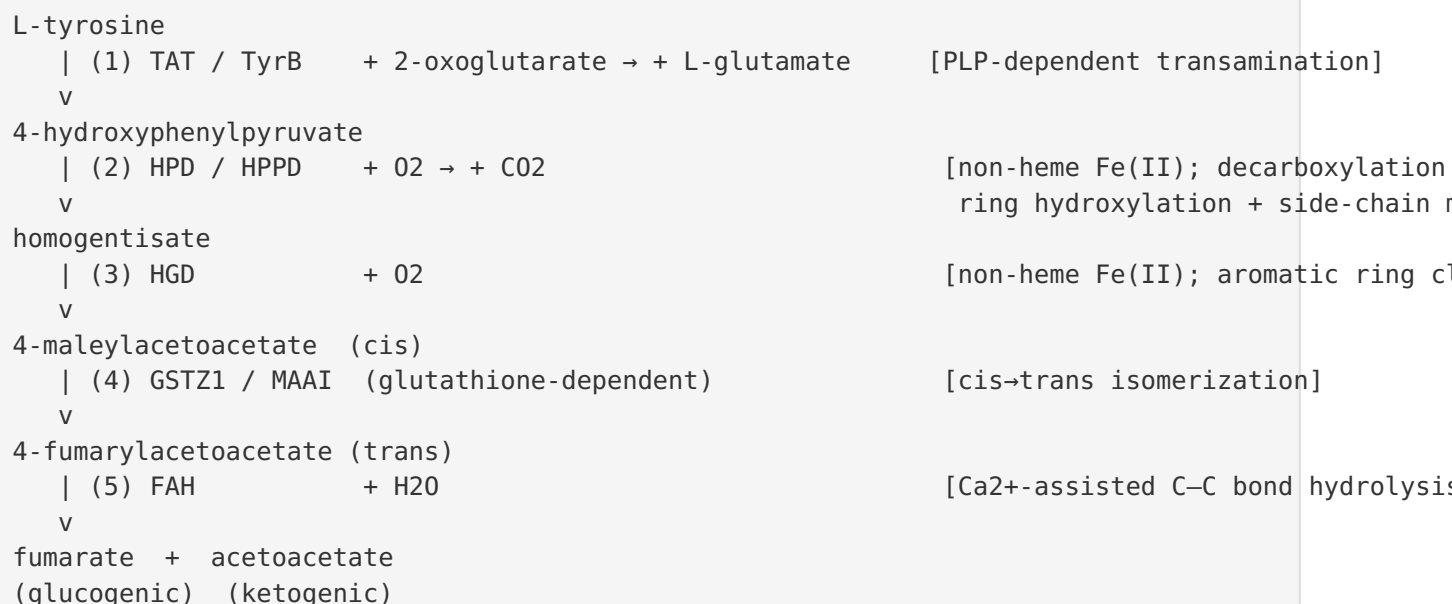
- **Anabolic diversion of homogentisate.** In plants, homogentisate is not primarily degraded but is redirected into biosynthesis of the essential redox metabolites **tocopherol and plastoquinone** (Dixon & Edwards, [PMID: 22980205](#)). This is a competing fate at the homogentisate branch point, not a step in the catabolic route.
- **The glutathione-transferase moonlighting functions of GSTZ1.** GSTZ1/MAAI is physically identical to glutathione transferase zeta-1 and participates in glutathione-dependent redox chemistry and xenobiotic (e.g., dichloroacetate) metabolism. These activities are distinct from its isomerase role in the pathway and are frequently conflated in the cancer-biology literature.

2.3 Competing definitions

The literature variously calls this the "tyrosine degradation pathway," the "phenylalanine/tyrosine catabolic pathway" (folding in PAH), or the "homogentisate pathway" (emphasizing the shared lower trunk used by multiple aromatic substrates). For the purposes of this review the tightest and most defensible definition is the **five-reaction tyrosine-to-fumarate/acetoacetate module** delimited at its top by the transaminase entry step and at its bottom by FAH hydrolysis.

3. Mechanistic Overview

3.1 The ordered reaction sequence



3.2 Obligatory, conditional, and accessory character of each step

- **All five steps are obligatory** for net conversion of tyrosine to fumarate + acetoacetate; there is no known bypass in animals. Genetic loss of any of steps 2, 3, or 5 produces a defined human disease (Section 6), demonstrating that the steps are non-redundant in vivo.
- **The entry transamination (step 1) is the conditional/regulated node.** In mammals, hepatic TAT is a classic glucocorticoid- and hormone-inducible enzyme; its expression is strongly controlled at the transcriptional and post-transcriptional level ([PMID: 29704219](#); [PMID: 22376142](#)). Bacteria use the broad-specificity aromatic aminotransferase TyrB instead. This step therefore sets pathway flux and is the principal point of physiological regulation.
- **The GSTZ1/MAAI isomerization (step 4) is chemically obligatory but biologically "quiet."** Maleylacetoacetate (cis) cannot be hydrolyzed by FAH, which requires the trans (fumaryl) isomer; the isomerization is therefore mechanistically required. Yet unlike its neighbors, GSTZ1/MAAI loss produces no unequivocal human disease (Section 6.3).

3.3 Molecular chemistry of each step

Step 2 — HPD/HPPD. A non-heme Fe(II)/O₂ dioxygenase that performs an unusually complex single reaction: oxidative decarboxylation of the 2-oxo acid coupled to aromatic ring hydroxylation and 1,2-migration of the side chain. It occurs across bacteria, plants, and animals and is the molecular target of the "triketone" bleaching herbicides and of the drug nitisinone (Trezza et al., [PMID: 38927403](#); [PMID: 40015842](#)).

Step 3 — HGD. A non-heme Fe(II) ring-cleaving dioxygenase that opens the aromatic ring of homogentisate to maleylacetoacetate. Human HGD is a hexamer; recent work has mapped a supramolecular oxygen-delivery architecture that channels O₂ to a deeply buried catalytic iron ([PMID: 42276179](#)), and directed-evolution studies show catalytic activity can be tuned by substitutions outside the catalytic pocket via hexamer stabilization ([PMID: 42665142](#)).

Step 4 — GSTZ1/MAAI. A glutathione-transferase-zeta-family enzyme that catalyzes glutathione-dependent *cis*→*trans* isomerization. It "is integral to the catabolism of the amino acids phenylalanine and tyrosine" (Stacpoole, [PMID: 37742772](#)); a zeta-class GST from *Arabidopsis* is a functional MAAI (Dixon & Edwards, [PMID: 22980205](#)).

Step 5 — FAH. Cleaves the C–C bond of fumarylacetoacetate using "a water molecule, activated by a catalytic His/Asp dyad, aided by a calcium ion that both chelates the enol acid form of the substrate and indirectly positions the water for nucleophilic attack at a carbonyl group" (Grogan, [PMID: 15934927](#)). FAH is the founding member of a large superfamily (Section 4).

4. Major Molecular Players and Active Assemblies

Step	Enzyme (families)	Fold / cofactor	Reaction	Key evidence
1	TAT (eukaryotic); TyrB (bacterial)	PLP-dependent aminotransferase	L-Tyr + 2-OG → 4-HPP + L-Glu	Glucocorticoid- inducible hepatic enzyme PMID: 29704219
2	HPD/HPPD	Non-heme Fe(II) dioxygenase (VOC- like)	4-HPP + O ₂ → HGA + CO ₂	Fe(II)/O ₂ mechanism; nitisinone/herbicide target PMID: 38927403
3	HGD	Non-heme Fe(II) ring- cleaving dioxygenase (hexamer)	HGA + O ₂ → maleylacetoacetate	Oxygen-tunnel architecture PMID: 42276179
4	GSTZ1/MAAI	GST-zeta fold (glutathione- dependent)	maleyl- → fumaryl- acetoacetate	Identical to GST zeta-1 PMID: 37742772
5	FAH	FAH superfamily (Ca ²⁺ /divalent-metal)	fumarylacetoacetate + H ₂ O → fumarate + acetoacetate	His/Asp dyad + Ca ²⁺ mechanism PMID: 15934927

A four-fold mosaic, not one gene family. The striking feature of the table above is that no two adjacent enzymes are homologous. The module was assembled by recruitment of distinct, pre-existing protein folds. This has an important corollary for evolutionary reasoning: one cannot infer the pathway's origin from a single phylogeny; each step has its own evolutionary history.

The FAH superfamily. FAH is the eponymous and best-characterized member of a large, ancient, and "largely unexplored protein superfamily" whose members share a C-terminal FAH fold with a divalent-metal active site coordinated by three conserved carboxylates (Brouns et al., [PMID: 18448118](#)). The superfamily is dominated by bacterial aromatic-degradation enzymes and sugar-acid dehydratases/hydrolases (e.g., KdaD, and the L-rhamnose-pathway hydrolase LRA6, [PMID: 36563174](#)), and includes the human mitochondrial paralog **FAHD1**, a bifunctional oxaloacetate decarboxylase / acylpyruvate hydrolase whose structure reveals a

flexible "lid" that folds over the active site upon ligand binding to complete a catalytic triad (PMID: 30348641). This superfamily context establishes that the terminal β -diketone-cleaving chemistry of the module is a specialization of a deep, mostly bacterial enzymatic lineage.

5. Evolutionary and Cell-Biological Variation

5.1 Across evolutionary lineages

- **Bacteria.** The homogentisate trunk is a widespread convergence point in aromatic-compound degradation, present in soil bacteria as a catabolic hub for multiple substrates (PMID: 24475028; PMID: 39730399). Bacteria use broad-specificity aromatic aminotransferases (TyrB) at the entry step.
- **Plants.** Plants possess the full lower pathway but principally use homogentisate anabolically (tocopherol/plastoquinone). When FAH is disrupted in *Arabidopsis*, toxic intermediates accumulate and cause spontaneous, light/short-day-dependent cell death that is suppressed by sugar and by lowering pathway flux (PMID: 23743712; PMID: 27097641) — a plant echo of the toxicity logic seen in human HT1.
- **Animals.** The pathway is a canonical hepatic (and renal) amino-acid catabolic route delivering glucogenic + ketogenic carbon. In hibernating bats, PAH, HPD, HGD, and FAH are co-upregulated and under stronger purifying selection, with conserved residues clustered at catalytically critical positions (PMID: 23620802), indicating physiological recruitment of the pathway for energy supply and detoxification.

5.2 Tissue, physiological state, and regulation

The entry transaminase is the regulated node. Mammalian hepatic TAT is strongly induced by glucocorticoids at the mRNA, protein, and activity level, integrating pathway flux with gluconeogenic and stress signaling (PMID: 29704219; PMID: 22376142). This places the top of the pathway under hormonal and nutritional control, whereas the lower trunk is largely constitutive and driven by substrate supply.

5.3 Disease-relevant expression changes

Both **HGD** and **GSTZ1** are downregulated in kidney renal clear cell carcinoma, where loss of tyrosine-metabolizing capacity contributes to metabolic reprogramming toward aerobic glycolysis (PMID: 35562974); GSTZ1 loss in hepatocellular carcinoma promotes metastasis via glucuronic-acid metabolism and TGF β signaling (PMID: 35979621). These illustrate that pathway components have acquired cell-biological roles beyond simple catabolism.

6. Constraints, Dependencies, and Failure Modes

6.1 Chemical ordering constraints

The sequence is enforced by substrate chemistry: each enzyme's product is the next enzyme's obligatory substrate (the "feeds-into" relationships stated in the brief are experimentally borne out). Two hard constraints deserve emphasis:

1. **The isomerization is not optional.** FAH acts only on the *trans* (fumaryl) isomer; the *cis* (maleyl) product of HGD must first be isomerized by GSTZ1/MAAI. There is no known route from maleylacetoacetate directly to products.
2. **Two O₂-dependent steps.** Both HPD and HGD require molecular oxygen at non-heme Fe(II) centers, making the pathway obligately aerobic. HGD in particular must solve an O₂-delivery problem to a buried iron (PMID: 42276179).

6.2 The severity gradient of pathway blocks

A unifying mechanistic principle emerged from this investigation: **the position of the metabolic block dictates both the severity of disease and the chemical identity of the toxic species.** The further downstream the lesion, the more electrophilic/reactive the accumulating intermediate.

Blocked enzyme	Disease	Accumulating species	Severity	Toxic mechanism
HPD	Tyrosinemia type III; hawkinsinuria	4-HPP / tyrosine	Mild–moderate	Elevated tyrosine; generally benign (PMID: 38927403)
HGD	Alkaptonuria	Homogentisic acid	Chronic, non-lethal	HGA oxidation → semiquinone-mediated ochronotic pigment; oxidative stress (PMID: 42665142; PMID: 41096940)
GSTZ1/MAAI	(no unequivocal human disease)	maleylacetoacetate / maleylacetone	Mild (experimental)	Maleylacetone cytotoxic; no succinylacetone (PMID: 12730618)
FAH	Hereditary tyrosinemia type 1	fumarylacetoacetate / succinylacetone	Severe, lethal	ALAD inhibition; liver failure; HCC (PMID: 17513424)

The molecular basis of HT1 toxicity — ALAD inhibition. The accumulating downstream metabolites are not merely inert waste; they are enzyme inhibitors. Succinylacetone inhibits human erythrocyte **δ -aminolevulinic acid dehydratase (ALAD / porphobilinogen synthase)** non-competitively with $K_i \approx 0.03 \mu\text{mol/L}$, and fumarylacetoacetate does so with $K_i \approx 0.06 \mu\text{mol/L}$ (Berger et al., PMID: 6652907). This produces the porphyria-like neurovisceral crises of HT1 — including newly described complications such as SIADH during acute decompensation (PMID: 41596311). That NTBC therapy "abolished" the near-complete inhibition of porphobilinogen synthase and normalized 5-aminolevulinate excretion (Lindstedt et al., PMID: 1383656) proves the accumulating intermediates are causal.

6.3 Why a GSTZ1/MAAI block is different

A pharmacological block of GSTZ1/MAAI (via dichloroacetate) in rats produces a **chemically distinct and milder signature**: dose-dependent urinary excretion of maleylacetoacetate and its decarboxylation product maleylacetone (cytotoxic to hepatocytes, $EC_{50} \approx 350 \mu\text{M}$, enhanced ~10-fold by homogentisate co-administration), but critically **succinylacetone was not**

detected and no morphological organ damage occurred despite measurable ALAD inhibition (Lantum et al., [PMID: 12730618](#)). Succinylacetone — the potent ALAD inhibitor and biomarker of HT1 — is the *reduced* analogue and is characteristic of a **downstream (FAH)** block, not a GSTZ1 block. This explains, at the level of chemistry, why "no clinical disease consequences are unequivocally attributable to inborn errors of this enzyme" (Stacpoole, [PMID: 37742772](#)).

6.4 The therapeutic logic of nitisinone (NTBC)

Because toxicity is generated downstream of HPD, the rational treatment for HT1 (an FAH deficiency) is to inhibit **HPD**, three steps upstream, starving the pathway of the toxic species. Nitisinone is a slow, tight-binding triketone that "preferentially binds to the complex of HPPD and Fe(II)" (visible absorbance at 450 nm) as a single dominant enol tautomer via a multi-step mechanism (pre-equilibrium $K_1 = 1.25 \text{ mM}$, $k_2 = 8.2 \text{ s}^{-1}$, $k_3 = 0.76 \text{ s}^{-1}$), consistent with iron chelation by the triketone (Kavana & Moran, [PMID: 12939152](#)). Clinically, NTBC decreases succinylacetone and protects against inflammation and oxidative damage ([PMID: 41003839](#)), and it is now being complemented by engineered-probiotic strategies to degrade tyrosine directly ([PMID: 42226462](#)). It converts lethal HT1 into a chemically milder, alkaptonuria/tyrosinemia-III-like state — again illustrating the severity gradient.

7. Controversies and Open Questions

1. **Why is GSTZ1/MAAI deficiency clinically silent?** No inborn error of GSTZ1 unequivocally causes disease in humans ([PMID: 37742772](#)), despite the isomerization being chemically obligatory. Possible explanations — residual non-enzymatic isomerization, redundancy, or the relative non-toxicity of maleyl-species versus succinylacetone ([PMID: 12730618](#)) — remain unresolved. This is arguably the single most interesting open question in the module.
2. **The exact HPD reaction coordinate.** HPD couples decarboxylation, hydroxylation, and side-chain migration in one active site; the precise oxygen-activation and migration mechanism remains an active area, informed by engineering studies that repurpose HPPD-type non-heme iron enzymes for new-to-nature chemistry ([PMID: 39260996](#); [PMID: 40259890](#)).

3. **O₂ access in HGD.** How molecular oxygen reaches the deeply buried catalytic iron of the HGD hexamer, and how AKU-associated variants perturb this, is only beginning to be dissected ([PMID: 42276179](#)). Directed-evolution work further shows activity is modulated by non-catalytic-pocket residues via oligomer stability ([PMID: 42665142](#)), complicating simple genotype–phenotype mapping in alkaptonuria.
 4. **Mechanism of ochronotic pigment formation in AKU.** The polymerization chemistry of homogentisic acid into ochronotic pigment — recently shown to proceed via a semiquinone radical, oxidative-coupling mechanism ([PMID: 41096940](#)) and to drive oxidative-stress-mediated autophagy/lysosomal failure in chondrocytes ([PMID: 42510554](#)) — is not fully defined and has no reversal therapy.
 5. **Cross-organism extrapolation.** Much mechanistic detail is drawn from bacterial and plant enzymes (e.g., *Arabidopsis* reconstitution, bacterial FAH-superfamily structures). While the fold and chemistry are conserved, quantitative kinetics, regulation, and toxicity thresholds should not be uncritically transferred between kingdoms.
 6. **Moonlighting and cancer biology.** HGD and GSTZ1 behave as tumor-suppressor-like metabolic genes in renal and hepatocellular carcinoma ([PMID: 35562974](#); [PMID: 35979621](#); [PMID: 36905252](#)). Whether these effects are downstream of pathway flux (e.g., fumarate levels) or of non-catabolic "moonlighting" activities is not settled.
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8. Key References

PMID	Contribution to this review
22980205	Reconstitution of a fully functional lower pathway (HGO+MAAI+FAH) in <i>E. coli</i> ; plant homogentisate diversion to tocopherol/plastoquinone
23743712	FAH hydrolyzes fumarylacetoacetate to fumarate + acetoacetate; plant FAH-loss cell death
17513424	FAH deficiency = tyrosinemia type 1; succinylacetone/succinylacetoacetate accumulation
38927403	HPD across bacteria/plants/animals; deficiency → tyrosinemia III and hawkinsinuria
37742772	GSTZ1/MAAI identity, function, and absence of a definitive human deficiency disorder
15934927	FAH catalytic mechanism (His/Asp dyad + Ca ²⁺); β-diketone cleavage diversity
18448118	FAH superfamily structural comparison and evolution
30348641	Human FAHD1 bifunctional FAH-superfamily paralog structure
36563174	Bacterial FAH-superfamily sugar-acid hydrolase (LRA6)
12939152	Nitisinone (NTBC) slow tight-binding inhibition of HPPD·Fe(II)
6652907	Succinylacetone/FAA inhibit ALAD (K _i 0.03/0.06 μmol/L) — molecular basis of HT1 toxicity
1383656	NTBC reverses porphobilinogen synthase inhibition in vivo
12730618	DCA-induced GSTZ1 deficiency → maleylacetone, not succinylacetone; milder phenotype
24475028	Homogentisate as a bacterial aromatic-catabolism convergence point
23620802	Coordinated pathway upregulation and purifying selection in hibernating bats
42276179	HGD hexamer oxygen-delivery architecture
42665142	HGD directed evolution; non-catalytic-pocket modulation; alkaptonuria
41096940	Semiquinone-mediated ochronotic pigment formation in AKU
29704219	Glucocorticoid regulation of hepatic TAT
35562974	HGD/GSTZ1 as metabolic-reprogramming biomarkers in renal carcinoma

9. Limitations and Knowledge Gaps

- **Organism mixing.** Several mechanistic anchors (pathway reconstitution, FAH-superfamily structures) are bacterial or plant; kinetic and regulatory parameters may differ in mammals.
 - **The isomerase paradox.** The absence of a human GSTZ1/MAAI disease is unexplained; whether this reflects redundancy, non-enzymatic isomerization, or low intermediate toxicity is unresolved.
 - **Entry-step coverage.** The transamination step (TAT/TyrB) was characterized mainly through its regulation and disease context; detailed structural/kinetic comparison of eukaryotic TAT vs bacterial TyrB was not pursued here.
 - **No direct experimental data were generated** in this investigation; conclusions rest entirely on synthesis of published literature (44 papers reviewed). Effect sizes and K_i values cited are from the original primary studies.
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10. Proposed Follow-up Actions

1. **Resolve the GSTZ1/MAAI silence.** Systematically screen human/animal GSTZ1-null models for subclinical maleylacetone accumulation and ALAD inhibition, and test whether non-enzymatic *cis*→*trans* isomerization sustains flux in vivo.
 2. **Quantitative flux/toxicity model.** Build a kinetic model of the five-step module parameterized with the measured K_i values for ALAD inhibition to predict metabolite accumulation and toxicity for each block position and for partial NTBC inhibition.
 3. **Structural completion.** Obtain high-resolution structures/mechanistic studies of HPD capturing the decarboxylation–hydroxylation–migration coordinate, and of the HGD oxygen tunnel with AKU variants.
 4. **Cross-kingdom kinetic comparison.** Directly compare mammalian, plant, and bacterial orthologs of each step under matched conditions to bound the validity of cross-organism extrapolation.
 5. **Therapeutic extension.** Evaluate combined upstream inhibition (NTBC) plus tyrosine-degrading engineered probiotics ([PMID: 42226462](https://pubmed.ncbi.nlm.nih.gov/42226462/)) to reduce tyrosine load and NTBC side-effect burden in HT1.
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Prepared as a commissioned review synthesis. All quantitative claims are attributed to the primary literature cited above; verbatim supporting quotations were validated against the source abstracts during the investigation.

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